SIH 26033

# FarmLink

## From Our Farms to Our Homes

A smarter way to connect farmers with nearby buyers, reduce avoidable costs, and plan deliveries better.

AGRICULTURE • FOODTECH • RURAL DEVELOPMENT

Student project proposal

# The Problem: A Price Gap Between Farm and Home

**An illustrative example**

**₹10/kg**

Farmer selling price

→

**₹30/kg**

Consumer buying price

**What widens the gap?**

Transportation • storage • handling • multiple intermediaries

**The difference is not always profit. Costs can build up between harvesting and the final sale.**

The challenge is limited buyer choice, limited price information and avoidable costs—not that every intermediary is unnecessary.

# Our Solution: FarmLink

## A digital bridge connecting farmers / FPOs with households, shops, hotels and bulk buyers.

01

### Farmers / FPOs

List crop, quantity, price and harvest date

→

02

### FarmLink

Match supply with nearby demand

→

03

### Buyers

Search, compare and place orders

→

04

### Shared delivery

Group orders and suggest practical routes

**Core idea: identify who needs produce, where it is needed, and how it can reach them efficiently.**

# How FarmLink Works

### 1 Register

Farmers, FPOs and buyers create profiles.

### 2 List harvest

Add crop, quantity, expected price and harvest date.

### 3 Find a match

Search nearby produce and compare supply with demand.

### 4 Confirm order

Agree on quantity, price and delivery timing.

### 5 Group delivery

Combine compatible orders where practical.

### 6 Track and learn

Share updates and measure outcomes.

**The first version can demonstrate the workflow with sample data before a real-world pilot.**

# Key Features for Farmers and Buyers

## Harvest Alerts

Farmers list expected harvests early and seek buyers before produce is ready.

## Nearby Buyer Discovery

Prioritize potential buyers within a practical delivery distance.

## Shared Orders

Combine compatible household, shop, hotel and bulk-buyer orders.

## Real Earnings Calculator

Estimate what remains after transport and packaging expenses.

## Price Comparison

Show available seller prices and quantities when reliable data exists.

## Order Updates

Confirm orders and communicate availability or delivery changes.

# Innovation: Harvest-to-Home Matching System

**FarmLink goes beyond listing produce: it connects expected future supply with local demand and shared transportation.**

01

### Harvest Alert

Farmer enters expected crop, quantity and approximate ready date.

02

### Demand Match

Check nearby households, shops, restaurants and bulk buyers.

03

### Shared Delivery Plan

Suggest a combined route when enough compatible orders exist.

**Example: 100 kg of tomatoes expected in 2 days; if demand is 70 kg, the farmer gets an early signal to seek more buyers.**

It supports better decisions but cannot guarantee that all produce will be sold.

# Smart Matching, AI and Delivery Planning

### Demand Prediction

Use historical orders, seasonal patterns and market information to estimate demand.

### Supply–Demand Matching

Compare expected harvest quantity with buyer requests and flag gaps.

### Route Suggestions

Use buyer locations, distance and vehicle capacity to suggest practical routes.

**Build in stages** Start with rule-based matching and delivery grouping. Add machine learning after collecting enough useful data.

Price comparison depends on reliable local market data being available.

# Technical Approach: Student-Friendly Prototype

| | |
|---|---|
| USER INTERFACE | HTML • CSS • JavaScript; React if required |
| BACKEND | Python with Flask |
| DATABASE | SQLite initially; PostgreSQL for expansion |
| SMART SERVICES | Python / Scikit-learn; OpenStreetMap; OSRM |

## Maps and routing

OpenStreetMap for geographic data; OSRM or a basic route algorithm.

## Deployment

A suitable free or low-cost hosting service for the prototype.

## Development principle

Keep the first version small, testable and easy to improve.

# Feasibility: Start Small, Prove the Idea

## First prototype scope

- Pilot one locality with a small group of users.
- Begin with tomatoes, onions and bananas.
- Let farmers list produce and buyers place sample orders.
- Demonstrate matching, order grouping and route suggestions.
- Use sample data first; seek local partners for later testing.

## Suggested build sequence

1. Farmer and buyer profiles
2. Produce listings and search
3. Order confirmation
4. Matching and earnings calculator
5. Delivery grouping and route demo

**Keep AI optional until enough useful data exists to evaluate it.**

# Challenges and How We Plan to Handle Them

| | | |
|---|---|---|
| **Digital comfort** | → | Simple interface and regional-language support |
| **Uncertain harvest** | → | Allow farmers to update expected quantity |
| **Order changes** | → | Confirm orders before arranging transport |
| **Perishable produce** | → | Prioritize time-sensitive products and delivery |
| **Limited AI data** | → | Start with simple rules and improve with real data |
| **Uneconomical small orders** | → | Set minimum combined quantities or show delivery cost |

# Expected Benefits and Impact

### For Farmers

More buyer options • clearer demand and price visibility • estimated earnings after delivery costs • less dependence on one sales channel.

### For Consumers

Nearby produce • easier price comparison • potentially lower distribution costs • clearer availability and harvest timing.

### For the Environment

Fewer unnecessary trips when orders are combined • potential reduction in food waste • more efficient vehicle use.

**These are intended benefits, not guaranteed outcomes. We will test them through a pilot.**

The goal is not to remove every intermediary, but to improve transparency, choice and coordination across the supply chain.

# How We Will Measure Whether FarmLink Works

| | | |
|---|---|---|
| 1 | **Farmer earnings** | Net amount after delivery expenses |
| 2 | **Buyer cost** | Price for the same quantity and quality |
| 3 | **Delivery efficiency** | Orders combined and distance travelled per order |
| 4 | **Food waste** | Amount of produce left unsold |
| 5 | **User adoption** | Number of buyers who return to use the platform |

**Compare pilot results with the existing selling method; report actual findings rather than assumed savings.**

# Pilot Plan and Practical Considerations

### Phase 1 • Prototype

Build listings, buyer search, order confirmation and the earnings calculator.

### Phase 2 • Demonstration

Use sample data to show matching, order grouping and route suggestions.

### Phase 3 • Pilot

If partnerships are available, test with real farmers and buyers in one locality.

## Important considerations

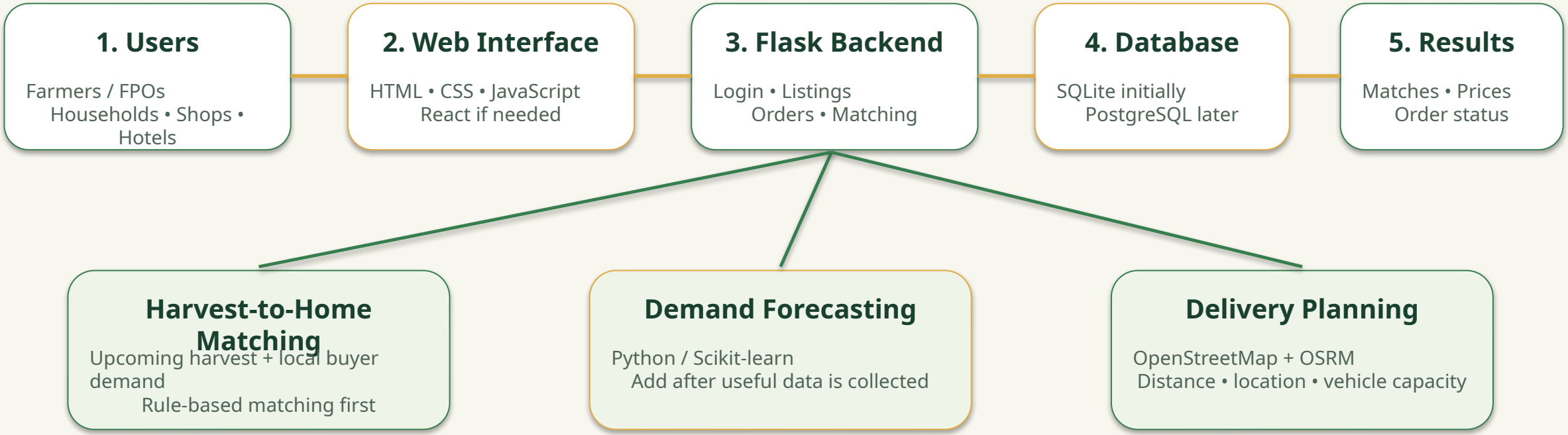
- Confirm orders before arranging transport.
- Show delivery costs before buyers commit.
- Support regional languages and changing harvest estimates.
- Treat AI predictions as estimates, not guarantees.

# Research and References

| | | |
|---|---|---|
| **e-NAM** | https://www.enam.gov.in/ | Electronic agricultural trading information |
| **AGMARKNET** | https://agmarknet.gov.in/ | Agricultural market prices and arrivals |
| **Ministry of Agriculture and Farmers Welfare** | https://agriwelfare.gov.in/ | Policies and farmer support initiatives |
| **OpenStreetMap** | https://www.openstreetmap.org/ | Geographic and mapping data |
| **OSRM** | https://project-osrm.org/ | Route calculation infrastructure |
| **Scikit-learn** | https://scikit-learn.org/ | Machine-learning tools |

Data availability and suitability must be checked for the selected locality.

# Technical Approach: FarmLink System Architecture




**OUTPUT: Better-informed selling decisions • grouped orders • suggested delivery routes • clearer estimated earnings**

Prototype stack: HTML/CSS/JavaScript • Python Flask • SQLite • OpenStreetMap / OSRM • Scikit-learn (later)

# FarmLink

## From Our Farms to Our Homes

**Why should a farmer struggle to find a buyer when people nearby may already need the same produce?**

Our aim is to match upcoming harvests with local demand, group nearby orders, and help farmers understand their actual earnings.

**Start small • Test with users • Measure results • Improve with evidence**

**Thank you**